

This project was undertaken with the primary **objective** of rigorously evaluating whether a personalized diet regimen is more effective in reducing average fasting glucose levels compared to a conventional, standard nutrition plan. To formally assess this, two opposing hypotheses were formulated: the null hypothesis (H0) posited that there would be no statistically significant difference in the reduction of fasting glucose between the personalized and standard nutrition plans, implying both would yield similar outcomes. Conversely, the alternative hypothesis (H1) proposed that personalized nutrition would lead to a significantly greater reduction in fasting glucose, suggesting its superior efficacy. This clear hypothesis framework guided the subsequent data collection and statistical analysis.

Prior to analysis, a crucial **data cleaning** phase was executed to ensure the reliability and integrity of the dataset. An initial inspection for missing values across key variables such as Patient_ID, Group, Glucose_Before, Glucose_After, and Glucose_Change confirmed that the dataset was complete with zero missing entries, which is ideal for direct analysis. Further scrutiny involved identifying and addressing outliers, which are data points significantly deviating from the general pattern. Three outliers were detected in the Glucose_Before measurements, and two outliers were found in the Glucose_After measurements. While the specific method of outlier handling (e.g., removal, transformation, or Winsorization) is not detailed, their identification indicates a thorough approach to data preparation.

Exploratory Data Analysis (EDA) provided foundational insights into the dataset's structure and characteristics. The Group Distribution confirmed an equal sample size, with 100 participants allocated to the Personalized nutrition group and 100 to the Standard nutrition group, ensuring a balanced comparison. The Summary Statistics by Group offered a quantitative overview of the glucose levels before and after intervention, as well as the change. For the Personalized group, the mean Glucose_Before was 158.44 mg/dL (median 158.10, standard deviation 13.63), which reduced to a mean Glucose_After of 138.11 mg/dL (median 137.90, standard deviation 13.67). This translated to a mean Glucose_Change (reduction) of 20.33 mg/dL (median 20.5, standard deviation 5.42). In contrast, the Standard group started with a mean Glucose_Before of 160.34 mg/dL (median 161.25, standard deviation 14.30), decreasing to a mean Glucose_After of 149.58 mg/dL (median 148.75, standard deviation 15.69). Their mean Glucose_Change was 10.75 mg/dL (median 10.4, standard deviation 6.19). These preliminary statistics already suggested a more pronounced glucose reduction in the Personalized group.

Before conducting parametric hypothesis tests, **normality testing** was performed using the Shapiro-Wilk test to ascertain whether the data distributions conformed to a

normal distribution, a key assumption for t-tests. For the Personalized group, Glucose_Before ($W=0.9899, p=0.6557$), Glucose_After ($W=0.9904, p=0.6990$), and Glucose_Change ($W=0.9858, p=0.3602$) all showed p-values greater than 0.05, indicating that their distributions were consistent with normality. Similarly, for the Standard group, Glucose_Before ($W=0.9776, p=0.0857$), Glucose_After ($W=0.9881, p=0.5126$), and Glucose_Change ($W=0.9898, p=0.6513$) also exhibited p-values exceeding 0.05, confirming their normal distribution. This successful normality check validated the appropriate use of t-tests for subsequent analyses.

The **hypothesis testing for within groups** involved assessing the effectiveness of each diet plan independently. A paired t-test was conducted for both groups to compare fasting glucose levels before and after the intervention. For the Personalized group, the paired t-test yielded a t-statistic of 37.5020 with an associated p-value of 0.0000. This highly significant p-value unequivocally indicated a statistically significant reduction in glucose levels within the Personalized group. Likewise, for the Standard group, the paired t-test resulted in a t-statistic of 17.3589 and a p-value of 0.0000, also demonstrating a statistically significant reduction in glucose levels within this group. These results confirmed that both personalized and standard diet plans were individually effective in lowering fasting glucose.

The core of the project's objective was addressed by the **hypothesis test for between groups**, specifically an independent t-test comparing the mean glucose reduction between the Personalized and Standard diet plans. As observed in EDA, the Personalized group achieved a mean glucose reduction of 20.33 mg/dL, while the Standard group achieved a mean reduction of 10.75 mg/dL. This translated to a difference in means of 9.58 mg/dL in favor of the personalized plan. The independent t-test produced a t-statistic of 11.6418 and a p-value of 0.0000. This extremely small p-value, far below the typical significance level of 0.05, led to the rejection of the null hypothesis. This outcome provided strong statistical evidence of a significant difference in glucose reduction between the two groups, confirming that the personalized diet plan led to a greater reduction.

Beyond statistical significance, the **effect size** was calculated using Cohen's d to quantify the practical magnitude of the observed difference. Cohen's d was found to be 1.6464, which is classified as a large effect size. This indicates that the difference in glucose reduction between the personalized and standard plans is not only statistically significant but also substantial and meaningful in a practical context. The mean reduction difference of 9.58 mg/dL with a pooled standard deviation of 5.82 further illustrates this large impact.

To provide a range of plausible values for the true population means, **confidence intervals** were constructed. The 95% confidence interval for the Personalized group's mean glucose reduction was calculated as [19.26,21.41] mg/dL. For the Standard group, the 95% confidence interval for their mean reduction was [9.52,11.98] mg/dL. Most importantly, the 95% confidence interval for the Difference in Mean Reduction (Personalized - Standard) was [7.96,11.21] mg/dL. Crucially, this interval does not include zero, reinforcing the conclusion that there is a significant difference between the two groups.

Results:

The comprehensive analysis culminated in the following key **results** and **insights**:

- Both the Personalized and Standard groups individually demonstrated statistically significant reductions in fasting glucose levels, as evidenced by their respective paired t-tests (p-values = 0.0000). This indicates that both dietary interventions were effective in lowering glucose.
- However, the Personalized group exhibited a demonstrably greater average reduction in fasting glucose compared to the Standard group. Specifically, the Personalized group's mean reduction was 20.33 mg/dL (with a 95% confidence interval of [19.26,21.41]), while the Standard group's mean reduction was 10.75 mg/dL (with a 95% confidence interval of [9.52,11.98]).
- The Difference in Mean Reduction between the Personalized and Standard groups was 9.58 mg/dL. This difference was found to be statistically significant (p-value = 0.0000), with a 95% confidence interval for the difference ranging from 8.43 mg/dL to 10.73 mg/dL.
- The calculated Effect Size (Cohen's d) of 1.65 signifies a large practical effect, emphasizing the substantial clinical relevance of the personalized diet's superior performance.

Conclusion:

Based on the robust statistical evidence, the project concludes that personalized diet plans significantly reduce fasting glucose more than standard nutrition plans. This supports the alternative hypothesis (H1).

Recommendations:

- **Integrate Personalized Nutrition into Clinical Practice:** Healthcare providers should consider incorporating personalized diet plans as a primary intervention for individuals seeking to manage or reduce fasting glucose levels, particularly in the context of chronic disease management.

- **Further Research on Personalization Factors:** Future studies should investigate the specific components or methodologies of personalized nutrition (e.g., genetic factors, microbiome data, lifestyle specifics) that contribute most significantly to glucose reduction to refine personalization strategies.
- **Long-term Efficacy Studies:** Conduct longer-term studies to assess the sustained impact of personalized diets on fasting glucose and other health markers, as well as patient adherence and quality of life.
- **Cost-Benefit Analysis:** Evaluate the cost-effectiveness of implementing personalized nutrition programs compared to standard care, considering both health outcomes and economic implications.
- **Patient Education and Support:** Develop comprehensive patient education materials and support systems to help individuals understand and adhere to personalized diet recommendations effectively.